Chapter 6: Light and Space

Comprehensive Study Notes & Worksheet

— TEACHER'S ANSWER KEY —

6.1 Reflection of Light

Light travels in extremely fast, **straight** lines. When light hits a surface and bounces off, we call this **reflection**.

- **The Law of Reflection:** The angle of **incidence** (i) is ALWAYS equal to the angle of **reflection** (r).
  - $i = r$
- **Key Terms for Ray Diagrams:**
  - **Incident Ray**: The incoming ray of light hitting the mirror.
  - **Reflected Ray**: The ray of light bouncing off the mirror.
  - **Normal**: An imaginary dotted line drawn at exactly **90**° (a right angle) to the mirror's surface at the exact point where the light hits.
  - **Angles:** Angles are ALWAYS measured between the light ray and the **Normal** (never between the ray and the mirror itself!).
- **Types of Reflection:**
  - **Specular** **Reflection:** Happens on perfectly smooth, shiny surfaces (like a mirror or still water). All light rays reflect in the same direction, creating a clear image.
  - **Diffuse** **Reflection:** Happens on rough surfaces (like paper or wood). Light rays scatter in many different directions, so no clear image is formed.
- **Images in a Plane (Flat) Mirror:**
  - They are **virtual** (the light rays don't actually meet behind the mirror, so the image cannot be projected onto a screen).
  - They are exactly the same size as the object.
  - They appear to be the same distance *behind* the mirror as the object is in front.
  - They are **laterally inverted** (left and right are swapped—this is why "AMBULANCE" is written backwards on the front of the vehicle!).

6.2 Refraction of Light

___Refraction___ is the bending of light when it passes from one transparent material (medium) into another.

- **Why does it happen?** Light travels at different ___speeds___ in different materials depending on their ___density___. It travels fastest in a vacuum, very fast in air, slower in water, and even slower in glass. When light hits a new medium at an angle, the change in speed causes the light ray to bend.
- **The Rules of Bending:**
  - **Less dense to more dense (e.g., Air into Glass/Water):** The light slows down and bends ___towards___ the normal.
  - **More dense to less dense (e.g., Glass/Water into Air):** The light speeds up and bends ___away from___ the normal.
- **Real-world examples of Refraction:**
  - **The Broken Pencil:** A straight pencil placed in a glass of water looks ___bent___ or broken at the surface because the light traveling from the underwater part of the pencil changes direction as it enters the air.
  - **Apparent Depth:** Swimming pools look ___shallower___ than they actually are. If you look at a coin at the bottom of a pool, refraction makes the coin appear higher up than its true position.
  - **Lenses:** Glasses, microscopes, and telescopes use curved pieces of glass (lenses) to deliberately refract light to magnify objects or correct vision.

6.3 Making Rainbows (Dispersion) & Seeing Colours

White light from the Sun isn't actually white—it's a **mixture** of all the colours of the rainbow combined!

- **Dispersion**: The splitting of white light into a spectrum of colours. We can do this using a triangular glass **prism**.
- **The Spectrum:** The colours of white light always appear in this exact order: **Red, Orange, Yellow, Green, Blue, Indigo, Violet** (ROYGBIV).
- **How it works:**
  - Different colours of light travel at slightly different speeds inside glass, meaning they refract (bend) at slightly different angles.
  - **Red light** bends the **least** (it is the fastest in glass).
  - **Violet light** bends the **most** (it is the slowest in glass).
- **Nature's Prisms:** Real rainbows form when millions of tiny water droplets suspended in the air after a rainstorm act as tiny prisms, refracting and dispersing the sunlight.
- **Why are objects coloured? (Absorption and Reflection):**
  - When white light hits a coloured object, the object **absorbs** some colours and **reflects** others.
  - A red apple looks red because it reflects red light into our eyes and absorbs all the other colours.
  - A perfectly **white** object reflects *all* colours.
  - A perfectly **black** object absorbs *all* colours (which is why wearing black on a sunny day makes you feel hotter!).

6.4 & 6.5 Galaxies and the Universe

Moving from light waves to the vastness of space!

- **What is a Galaxy?** A massive, spinning collection of billions of stars, dust, gas, and dark matter, all held together by **gravity**.
- **Types of Galaxies:**
  - **Spiral** **Galaxies:** Have a central bulge and twisting arms made of stars and gas (e.g., our **Milky Way**).
  - **Elliptical** **Galaxies:** Shaped like giant ovals or spheres. They mostly contain older stars and have less dust and gas.
  - **Irregular** **Galaxies:** Have no defined shape, often because they have collided with other galaxies.
- **The Milky Way:** This is our home galaxy. Our Solar System is located on one of its outer spiral arms, called the Orion Arm. If you look at the night sky in a very dark place, you can see the dense center of our galaxy as a milky band of light.
- **The Universe:** The universe is everything that exists. It contains billions of different galaxies separated by vast amounts of completely empty space (a vacuum). Scientists believe it began expanding from a single point during the **Big Bang** around 13.8 billion years ago.
- **Light Years:** Because space is so enormous, measuring in kilometers is impossible (the numbers would be too long). Scientists use **light-years** to measure distance.
  - A light-year is the **distance** that light travels in one Earth year (about 9.46 trillion kilometers).
  - *Fun Fact:* The nearest star to us (other than the Sun) is Proxima Centauri, which is about 4.2 light-years away. That means the light we see from it today actually left the star 4.2 years ago!

6.6 Rocks in Space

Not everything in space is a huge star or a round planet. There is a lot of rocky and icy debris leftover from the formation of the solar system 4.6 billion years ago.

- **Asteroids**: Lumps of solid rock and metal.
  - They have irregular, bumpy shapes.
  - Most are found in the **Asteroid Belt**, a huge ring of space rubble orbiting the Sun between the orbits of **Mars** and **Jupiter**.
  - *Fun fact:* It is widely believed that a massive asteroid impact on Earth 66 million years ago (the Chicxulub crater) caused the extinction of the dinosaurs.
- **Comets**: Often called "dirty snowballs," these are made of loose rock, dust, and frozen gases/ice.
  - They have highly oval-shaped (elliptical) orbits that take them to the very freezing edges of the solar system and then close to the Sun.
  - As they get close to the heat of the Sun, the ice melts and vaporises into gas. This forms a glowing **coma** (cloud) around it and a long **tail** that can stretch for millions of kilometers.
  - The tail *always* points **away** from the Sun because it is pushed by solar winds.
  - *Famous Example:* Halley's Comet, which is visible from Earth every 75–76 years.
- **Meteoroids, Meteors, and Meteorites:** (The 3 M's of Space Rocks)
  - **Meteoroid**: A small piece of rock or metal travelling completely freely through space.
  - **Meteor**: A meteoroid that enters Earth's atmosphere. It travels so fast that air friction heats it up, causing it to burn and create a bright streak of light. We commonly call these **"shooting stars"**.
  - **Meteorite**: If a meteor is large enough, it won't burn up completely. The chunk of rock that actually survives and crashes into the Earth's surface is called a meteorite.

Chapter 6: Student Worksheet

Read the study notes carefully, then answer all 50 questions below.

Section A: Multiple Choice Questions (Circle the correct answer)

1. How does light travel?
   - A. In wavy lines
   - B. **In straight lines**
   - C. In circles
   - D. It teleports
2. According to the Law of Reflection, if the angle of incidence is $40°$, what is the angle of reflection?
   - A. $20°$
   - B. $50°$
   - C. $40°$
   - D. $90°$
3. Where is the angle of incidence measured?
   - A. Between the incident ray and the mirror
   - B. **Between the incident ray and the normal**
   - C. Between the reflected ray and the normal
   - D. Between the reflected ray and the mirror
4. Which type of reflection happens on a completely smooth, shiny surface like still water?
   - A. Diffuse reflection
   - B. Virtual reflection
   - C. **Specular reflection**
   - D. Refractive reflection
5. Why does the word "AMBULANCE" appear backwards on emergency vehicles?
   - A. Due to diffuse reflection
   - B. Due to refraction in the air
   - C. **Due to lateral inversion in plane mirrors**
   - D. So the driver can read it
6. What happens to light when it travels from air into a glass block?
   - A. It speeds up and bends towards the normal
   - B. It speeds up and bends away from the normal
   - C. **It slows down and bends towards the normal**
   - D. It slows down and bends away from the normal

7. Which phenomenon makes a swimming pool look shallower than it actually is?
   A. Reflection
   B. **Refraction**
   C. Dispersion
   D. Absorption
8. What equipment is used to split white light into a spectrum in a laboratory?
   A. A plane mirror
   B. A glass block
   C. **A triangular glass prism**
   D. A magnifying lens
9. Which colour of light travels the SLOWEST inside a glass prism?
   A. Red
   B. Green
   C. Yellow
   D. **Violet**
10. Why does a green leaf appear green under sunlight?
   A. It absorbs green light and reflects all other colours.
   B. **It reflects green light and absorbs all other colours.**
   C. It creates its own green light.
   D. It refracts green light away from the eye.
11. Which shape accurately describes our home galaxy, the Milky Way?
   A. Elliptical
   B. Irregular
   C. **Spiral**
   D. Spherical
12. What does a "light-year" measure?
   A. The speed of a star
   B. Time
   C. **The distance light travels in one year**
   D. The brightness of a galaxy
13. Where is the main Asteroid Belt located?
   A. Between Earth and Mars
   B. **Between Mars and Jupiter**
   C. Past the orbit of Neptune
   D. Between Venus and Earth

14. Why does a comet have a glowing tail?
    A. It is burning up in Earth's atmosphere.
    B. **The ice melts and vaporises into gas near the Sun.**
    C. It reflects the light of nearby stars.
    D. It is made of fire.
15. What do we call a piece of space rock that survives the atmosphere and hits Earth's surface?
    A. Meteoroid
    B. Meteor
    C. **Meteorite**
    D. Asteroid

Section B: Fill in the Blanks

1. When light hits a surface and bounces off, we call this **reflection**.
2. The imaginary dotted line drawn at exactly 90° to a mirror's surface is called the **normal**.
3. **Diffuse** reflection happens on rough surfaces like wood, scattering light in many directions.
4. An image in a plane mirror cannot be projected onto a screen because it is a **virtual** image.
5. The bending of light when it passes from one transparent medium to another is called **refraction**.
6. Light travels fastest in a **vacuum**.
7. When light travels from water into air (more dense to less dense), it bends **away** from the normal.
8. The splitting of white light into a spectrum of colours is known as **dispersion**.
9. The colours of the visible spectrum in order are Red, Orange, Yellow, Green, Blue, Indigo, and **Violet**.
10. A perfectly **black** object absorbs all colours, which is why it heats up faster on a sunny day.
11. A **galaxy** is a massive collection of billions of stars, dust, and gas held together by gravity.
12. Scientists believe the universe began expanding from a single point during the **Big Bang**.
13. Comets are often described as "dirty **snowballs**" because they are made of loose rock, dust, and ice.
14. A comet's tail always points **away** from the Sun because it is pushed by solar winds.
15. The common name for a meteor burning up in the Earth's atmosphere is a "**shooting** star".

Section C: Short Answer & Explanation

1. State the Law of Reflection.

   The angle of incidence is always equal to the angle of reflection ($i = r$).

2. Explain the difference between the incident ray and the reflected ray.

   The incident ray is the incoming light hitting a surface, while the reflected ray is the light bouncing off.

3. List three properties of an image formed in a plane (flat) mirror.

   It is virtual, laterally inverted, and the exact same size as the original object.

4. Explain clearly why a straight pencil looks bent or broken when placed in a glass of water.

   Light travels at different speeds in water and air. Light from the underwater pencil refracts (bends) as it enters the air.

5. Explain why a violet light ray bends more than a red light ray when passing through a glass prism.

   Violet light travels slower through the glass than red light does, causing it to bend (refract) at a sharper angle.

6. How do real rainbows form in nature after a rainstorm?

   Millions of water droplets suspended in the air act as tiny prisms, refracting and dispersing white sunlight into a spectrum.

7. If you wear a white t-shirt on a sunny day, why do you feel cooler than someone wearing a black t-shirt?

   A white t-shirt reflects all colours of light (and heat), whereas a black t-shirt absorbs all colours, trapping the heat.

8. Name the three main shapes/types of galaxies found in the universe.

   Spiral, Elliptical, and Irregular galaxies.

9. Why do scientists use "light-years" to measure distances in space instead of kilometers?

   Space is so vast that using kilometers would result in numbers that are too long and impractical to calculate with.

10. Proxima Centauri is 4.2 light-years away. If that star exploded today, when would we see it happen on Earth? Explain.

    We would see it in 4.2 years, because it will take exactly 4.2 years for the light from the explosion to travel to Earth.

11. What is the difference between an asteroid and a comet in terms of what they are made of?

    Asteroids are made of solid rock and metal, while comets are made of loose rock, dust, and frozen gases/ice.

12. Describe the shape of a comet's orbit around the Sun.

    Comets have highly elliptical (oval-shaped) orbits that bring them very close to the Sun and then far out to the edge of the solar system.

13. Define what a meteoroid is.

    A small piece of rock or metal travelling completely freely through space.

14. Why does a meteor create a bright streak of light when it enters Earth's atmosphere?

    It travels so incredibly fast that friction with the air particles heats it up, causing it to burn and glow brightly.

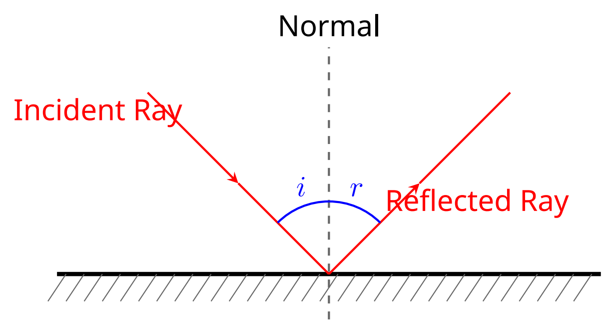
15. What is the scientific theory regarding the extinction of the dinosaurs 66 million years ago?

    A massive asteroid impacted the Earth, altering the global climate drastically and leading to a mass extinction.

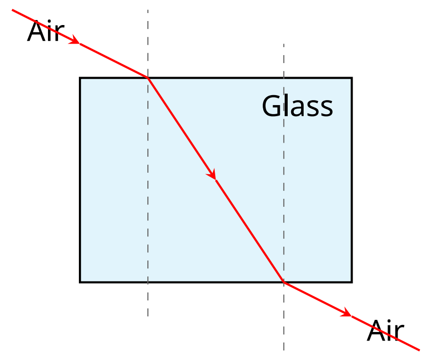
Section D: Drawing & Sketching Challenges (TEACHER KEY)

Use a pencil and ruler where necessary to complete the following diagrams based on your notes.

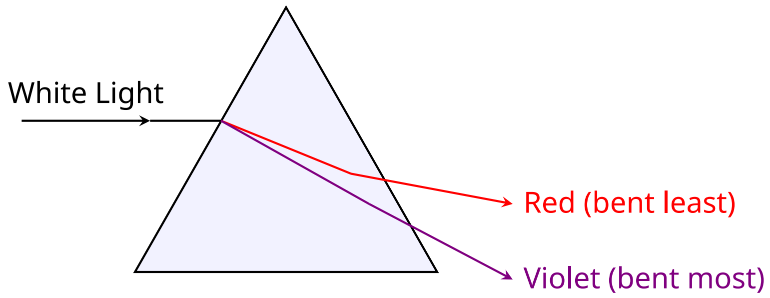
1. **Reflection Diagram:** Sketch a flat mirror. Draw an incident ray hitting the mirror, the normal line, and the reflected ray. Label the angle of incidence (i) and the angle of reflection (r).



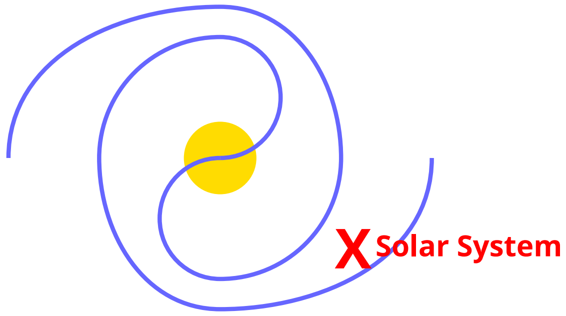
2. **Refraction Diagram:** Draw a rectangular glass block. Show a ray of light traveling from the air, entering the glass block, and then exiting back into the air. Show how the light bends at both boundaries.



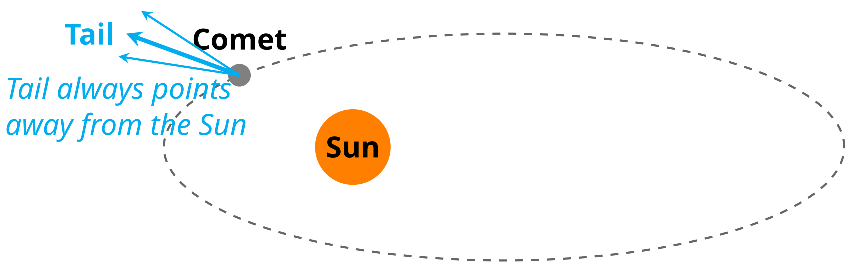
3. **Dispersion Diagram:** Draw a triangular glass prism. Show white light entering the prism and splitting. Label the colours that bend the **most** and the **least** as they exit.



4. **Galaxy Sketch:** Sketch the shape of a **spiral galaxy** (like our Milky Way) viewed from above. Place an 'X' to mark the approximate location of our Solar System on one of the outer arms.



5. **Comet Orbit:** Sketch the Sun in the center. Draw the highly elliptical orbit of a comet around the Sun. Draw the comet itself at a point close to the Sun, ensuring you draw the tail pointing in the correct direction.



End of Answer Key.